Sebastian Dziadzio

I am an AI researcher specializing in continual pretraining and benchmarking agentic systems. I have ten years of experience in machine learning research across startups, industry labs, and academia. I published at NeurIPS, CVPR, ACL, and ICCV, with 750+ citations, and hold two patents.

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Email
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GitHub

EXPERIENCE

2022–Now **Doctoral Researcher, Bethge Lab, University of Tübingen** — Tübingen, Germany

I developed methods and benchmarks for continually updating multimodal foundation models and evaluating them on open-ended capabilities. More recently, I built agentic benchmarks for inferring mechanisms from behavior. Anticipated graduation: December 2026.

2019–2022 **Scientist, Microsoft Research, Mixed Reality & AI Lab** — Cambridge, United Kingdom

I owned and extended a parametric 3D face model used across multiple projects. I built the training, evaluation, and visualization infrastructure, and led a research effort that reduced the average fitting error by half.

2018–2019 **AI Resident, Microsoft Research** — Cambridge, United Kingdom

I conducted research on GANs for novel view synthesis and on deep RL for video games.

2016–2018 **Data Scientist, Cliqz** — Munich, Germany

I built an information retrieval model combining LSTMs and CNNs.

2015–2016 **Software Engineer, Nokia Networks** — Kraków, Poland

I implemented LTE control plane features and designed the test suites.

EDUCATION

2022–Now **PhD, Computer Science, University of Tübingen** — Tübingen, Germany

ELLIS PhD Program (3% acceptance rate), IMPRS-IS (5%)

2014–2016 **MSc, Computer Science, AGH University** — Kraków, Poland

2010–2014 **BSc, Acoustical Engineering, AGH University** — Kraków, Poland

EXPERTISE

Research areas

Benchmarking and evaluation, continual multimodal pretraining, model merging, agentic systems

Technical skills

Python, PyTorch, Docker, SLURM, DeepSpeed, vLLM

Languages

Polish (native), English (fluent), German (intermediate), Spanish (conversational)

SELECTED PUBLICATIONS

* denotes equal first contribution

RevengeBench: Reverse Engineering Code-Space Policies from Behavioral Experiments
Under review, NeurIPS 2026
B. Rahmani*, S. Dziadzio*, J. Strüber*, S. Hernández-Gutiérrez, M. Bethge.

ONEBench to Test Them All: Sample-Level Benchmarking Over Open-Ended Capabilities
ACL 2025
A. Ghosh*, S. Dziadzio*, A. Prabhu, V. Udandarao, S. Albanie, M. Bethge.

How to Merge Your Multimodal Models Over Time?
CVPR 2025
S. Dziadzio*, V. Udandarao*, K. Roth*, A. Prabhu, Z. Akata, S. Albanie, M. Bethge.

A Practitioner's Guide to Continual Multimodal Pretraining
NeurIPS 2024
K. Roth*, V. Udandarao*, S. Dziadzio, A. Prabhu, M. Cherti, O. Vinyals, O. Hénaff, S. Albanie, M. Bethge, Z. Akata.

Disentangled Continual Learning: Separating Memory Edits from Generalization
CoLLAs 2024
S. Dziadzio, Ç. Yıldız, G. v/d Ven, T. Trzciński, T. Tuytelaars, M. Bethge.

Full-Body Motion from a Single Head-Mounted Device
ICCV 2021
A. Dittadi, S. Dziadzio, D. Cosker, B. Lundell, T. Cashman, J. Shotton.

Fake It Till You Make It: Face Analysis in the Wild Using Synthetic Data Alone
ICCV 2021
E. Wood, T. Baltrušaitis, C. Hewitt, S. Dziadzio, T. Cashman, J. Shotton.

PATENTS

Controllable Image Generation
US-11748932, 2023
M. Kowalski, S. Garbin, M. Johnson, T. Baltrušaitis, M. De La Gorce, V. Estellers, S. Dziadzio.

Computing Photorealistic Versions of Synthetic Images
US-11354846, 2022
S. Garbin, M. Kowalski, M. Johnson, T. Baltrušaitis, M. De La Gorce, V. Estellers, S. Dziadzio, J. Shotton.

ACADEMIC SERVICE

Organization
ML in PL Conference 2022–2026 (organizer and team lead), ML Summer School 2022–2026

Reviewing
NeurIPS, ICML, ICLR, CVPR

Supervision
One MSc and one PhD student